

Memory and Misinformation Effect: Exploring the Influence of Different Types of Perceptual Misinformation on Memory

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Abstract

We are exposed to a broad range of information daily, unfortunately, not all of it is accurate nor correct, which represents a threat to our ability to gauge the accuracy of our own memories. The misinformation effect has been widely researched due to its importance and its ability to alter our original memories. This resulted in a robust body of research aiming to delve into the intricacies of the impact of misinformation it potentially has. Throughout this project, we aim to explore the influence of different types of perceptual misinformation on the memory using the within subject experiment design. In order to do so, we also analyzed the reaction time to find out whether the participants would take longer in attempting to recall the original information.

Keywords: Memory misinformation, False memory

Introduction

We all might think that our memories are accurate and reliable. However, in reality they are not as reliable as we think they are. As Elizabeth F. Loftus, prominent psychologist specializing in misinformation effect and false memory said “Memory is not a recording device. It is like a wikipedia page and can be changed by you or someone else”. Human brain has the tendency to adapt and modify memories in response to new information. Exposure to misinformation can lead to altering our original memory. There has been several research done in the area of false memory creation. Loftus (1999) and Garry et al. (2005) have demonstrated the formation of false memories in individuals by showing participants altered (unreal) images of events that had never occurred in their lives, such as getting lost in a mall and a ride of a hot air balloon, respectively. These images led participants to falsely recall these events and vividly describe them as if they had experienced them. Thus, there is an influence of visual misinformation and its impact on the memory. Roesch et al. showed that changing the words of questions impacts the way individuals answer. The experiment included showing a car accident scene, followed by asking questions about what they saw. The study found out that changing some particular words leads the participants to recall things which were originally not in the scene. Karanian et al. (2020) explored the three condition types: no warning, warning before and warning after the misinformed narration. Previous work has shown that participants do assimilate misinformation and change their original memory. Extending these works we intend to explore the influence of different types of misinformation which in here we have categorized as colour, location and replace.

Methods

Participants

Our experiment had a total of 45 participants, where the majority of them were Hanyang university students. The age of participants ranged from 18 to 27 years. The majority of the participants had English as their second language. Recruitment was conducted by sharing a Google form link to acquaintances and known contacts. Initially, we had 48 participants, however due to some missing data points we had to exclude their responses from the analysis. The experiment was within subject type and each participant was shown all the three images.

Apparatus and Stimuli

Images

The images used in this experiment were created specifically for this research study. All three images were created using ToonyTool, an online tool for generating cartoon style images (as shown in Figure 1). The design and complexity of the images were made keeping in mind the duration for which they were shown (5 seconds). Each image depicted a basic, everyday scene to ensure easy understanding for the participants. The objects/props used in each image were simple and different along all three, to make sure that all the three scenes are unique.

The brief descriptions of each image are as follows:

Image 1: This picture depicted a park scene.

Image 2: This picture depicted an outdoor restaurant scene.

Image 3: This picture depicted an office scene.

Audio

The audio accompanying each image was scripted and recorded very carefully. The important highlight of this experiment is that we created three different types of misinformation, misinformation about colour, misinformation about location and misinformation about replacing an object. Each image audio had only one of these

misinformation. The main aim (use) of audio was to precisely describe the entire image which was shown while altering one feature in it, i.e incorporating misinformation. This approach was applied to make sure that the participants could rely on the memory of the image they saw, with one exception which was the misinformation. The description of each image was first designed and then the text was converted into a voice using an online application named tts speech maker. Considering that most of the participants had English as their second language, the audio speed was adjusted to a slower speed (0.9x) and an easy and clear accent was used for better understanding.

The brief description of the type of misinformation incorporated into each audio is as follows:

Audio 1: Type of Misinformation- Colour

This audio described the whole scene of the park image but included misinformation by describing the color of the ball as blue, while in the image the original colour is yellow.

Audio 2: Type of Misinformation- Location

This audio described the whole scene of outdoor restaurant image but included misinformation by changing the location of the dog to the right side (near the man), while originally it is placed on the left side (near the women)

Audio 3: Type of Misinformation- Replace

This audio described the whole scene of the office, but included misinformation by replacing 'holding cup of coffee' to 'holding files'.

Design and Procedure

The entire experiment was designed and performed on PsychoPy. Our variables were as follows.

Dependent Variable: Accuracy of memory recall (whether they were misinformed or not)

Independent Variable: Three types of misinformation

1. Colour 2.Location 3.Replace

Design of questions:

There were 5 multiple choice questions (MCQ) for each image. In these 5 questions, only one question was about the misinformation. In total, each participant answered 15 questions (3image*5 questions = 15). We can divide these 15 questions into 12 general baseline questions(no misinformation) and 3 misinformation questions.

Design of presentation order:

We had created three different image order groups to reduce any kind of biases related to the presentation order. In each order, the first image was different. The idea was to create three groups, each group consisting of 15 participants.

Group 1: Order of presenting the images: 123

Group 2: Order of presenting the images: 213

Group 3: Order of presenting the images: 312

The focus was to ensure that each of the three images appeared first in one of the groups, so as to balance out order effects and reduce any possible carryover effect.

Procedure:

Flow of the experiment is explained as follows (also shown in Figure 1):

The image was shown for 5 seconds followed by a distractor task of clicking on boxes that popped on the screen; this was done in order to prevent rehearsal of the seen image. After that, participants heard an audio about the image they saw, the duration of audio was about 30 seconds. Followed by that, there were 5 multiple types of questions (MCQ) to be answered by them.

This procedure was repeated a total of 3 times, i.e for 3 images. The instructions were clearly shown on the screen one by one throughout the experiment. We made sure that participants answered the questions solely on the image they saw, by instructing 'Answer the following questions based on the IMAGE you saw' this was done in order to strengthen our results of incorporating misinformation.

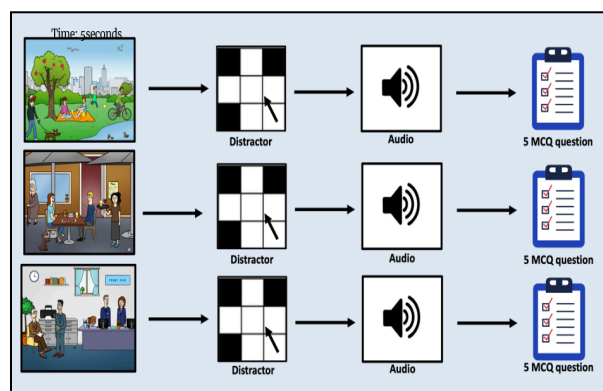


Figure 1: Procedure of the experiment.

Analysis

We conducted analyses focusing primarily on three key aspects:

1. Overall Analysis of Misinformation Type

Overall, we wanted to check if there is a significant difference in the number of misinformed and not misinformed participants across the types of misinformation, i.e color, location and replace to determine if a particular type of misinformation is more prone to change the original memory.

2. Analysis of the Initially Shown Image

We wanted to determine if there is a difference in the performance amongst the first shown picture in each group in the 3 different orders we made.

3. Reaction Time Analysis

We wanted to find out whether participants took longer time on the misinformed questions (3 questions) as compared to the normal/basic questions (12 questions). We hypothesized that the questions having misinformation are a little tricky and would require participants to think more carefully, hence a longer response time.

Baseline Performance Analysis:

The performance on these general questions serves as a baseline to compare against the misinformation questions. As seen in Figure 2; under normal conditions, most participants have a high accuracy rate. The majority of participants answered the general questions correctly, indicating that participants generally understood the images and audio descriptions. However, there are some incorrect responses as well, but these can be treated as general human error, i.e some participants might have faced difficulty in recalling specific details or might have faced some confusion.

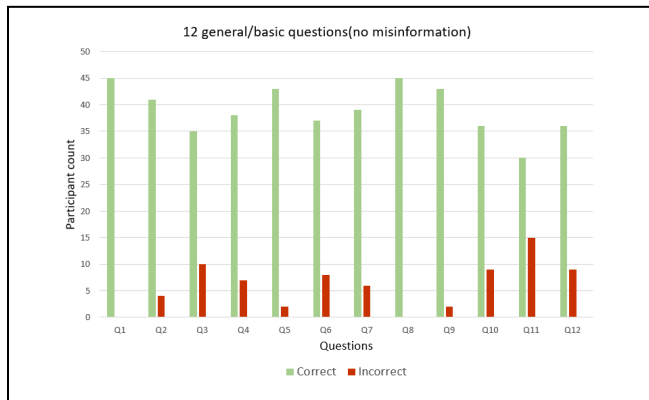


Figure 2: Baseline performance questions.

The results and conclusions for each of these analyses will be presented in their respective sections.

Results

Result 1

The plotted graph and results are displayed in Figure 3. From the graph we can observe that there are 16 participants who got misinformed for the type 'colour' and 'location', whereas there are 21 participants who got misinformed for the type 'replace' (each out of a total of 45). We conducted the 'Chi square test' to check for the significance level of difference among the three types. The p-value obtained was 0.460 ($p\text{-value} > 0.05$) and the calculated chi squared value (X^2) is lesser than tabular chi square value (figure 4). This shows that there is no significant difference among the different types of misinformation (color, location and replace). Thus, we can not reject the null hypothesis.

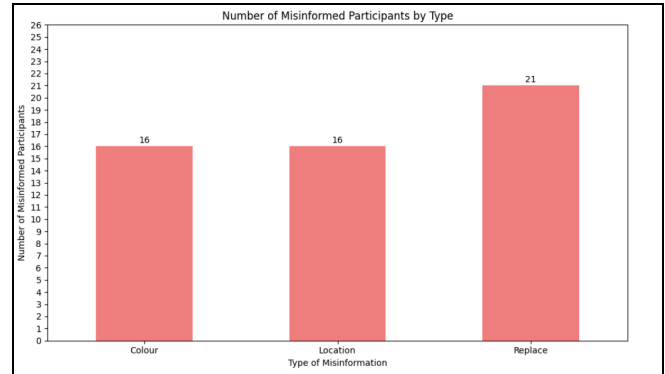


Figure 3: Misinformed participants by type.

Contingency Tables ▼

Contingency Tables ▼

Type	Not misinformed or Misinformed		Total
	0	1	
Colour	29	16	45
Location	29	16	45
Replace	24	21	45
Total	82	53	135

Chi-Squared Tests

	Value	df	p
X^2	1.553	2	0.460
N	135		

Figure 4: Chi Square test for the different types of misinformation questions.

Result 2

As seen in the Figure 5 when the image 1(color) was shown first in the order 123 there were 8 not misinformed participants and 7 misinformed participants, when image 2 (location) was shown first in the order 213 there were 11 not misinformed participants and 4 misinformed participants and finally when image 3(replace) was shown first, there were 7 not misinformed participants and 8 misinformed participants. For analyzing the difference among the three first shown images we ran a Chi square test. The results came out to be that p value was 0.306 (shown in Figure 6). The analysis revealed that there is no significant difference in order of showing an image first in the trial.

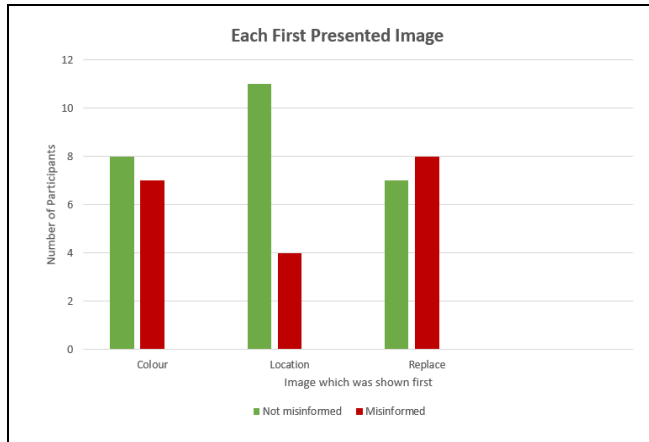


Figure 5: The number of participants per misinformation level in the initially shown images in each order.

Contingency Tables ▼

Contingency Tables ▼

Type	Not misinformed or misinformed		Total
	0	1	
Colour	8	7	15
Location	11	4	15
Replace	7	8	15
Total	26	19	45

Chi-Squared Tests

	Value	df	p
X ²	2.368	2	0.306
N	45		

Figure 6: Chi square test showing the number of participants per misinformation level in the initially shown images in each order.

Result 3:

As we were interested in exploring whether the participants would take longer to answer the misinformation questions, we conducted a paired t-test between the reaction time of the averaged 3 misinformation questions and the averaged 12 remaining baseline questions as shown below(Figure 7).

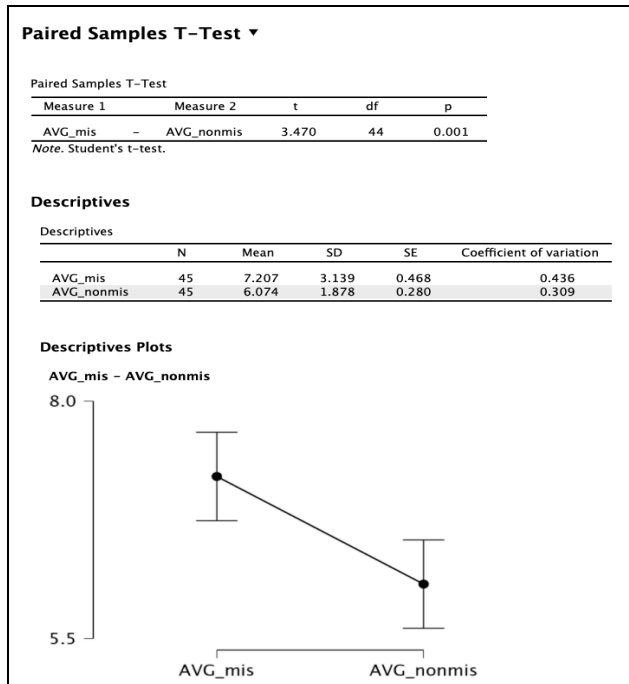


Figure 7: Results of paired t-test between the reaction time average of the misinformation questions and the average of the baseline questions.

As the p-value was 0.001 which is smaller than the significance level ($p < 0.05$), we can reject the null hypothesis and state that there is a significant difference in terms of the time taken to answer the misinformation questions as compared to the baseline questions.

Conclusion

Conclusion 1

The conclusion for first analysis shows that there is no significant difference in the type of misinformation. We can not say whether a specific type of misinformation leads to higher rate of misinformation incorporation. Hence, not being able to reject the null hypothesis.

Conclusion 2

The finding shows that the order in which these three images were presented did not lead to significant variations in participants' performance of incorporating the misinformation. We can not clearly determine if a particular first shown image could have a better correct recall memory. In sum, the initial presentation order of the three images did not affect how participants responded to the questions.

Conclusion 3

As we could reject the null hypothesis stating that there is no significant difference in the time taken in answering the misinformation and baseline questions, hence, we could conclude that participants needed more time to recall the original seen information and compare it to the misinformation they perceived through the narration.

Limitations and Future Work

While we carefully designed the whole experiment, there are few limitations to it. Firstly, we collected data of limited participants. We could only run the experiment on 45 individuals. Which leads to the drawback of not generalizing. The results of this study can not be generalized to a new group of people. If only we could have collected more data, our results could have come out more statistically powerful. One more limitation is that we only presented images, however this study can be extended to include videos which would provide longer visual stimuli .

This experiment has further potential for enhancement, especially in the area of different types of misinformation. This study can be taken to the next level by researching the particular reasons behind misinformation and its effects on memory. While this experiment was limited to showing images, it could be extended to include videos which have more information. Future research can also explore, tracking the eye movement of participants while they view the images, which could offer insights about the parts of the image an individual pays attention to. Additionally, examining the effects of misinformation across different age groups could be a possible interesting research area by analyzing the different impact misinformation has on the memory.

Contributions

Throughout this research, we attempted to contribute to the area of misinformation effect on memory recall methodologically and theoretically. First, we adopted the within subject type in this experiment where all the participants were shown the three images. The purpose behind this is to have a comprehensive examination of how each participant would respond to the different types of misinformation. Secondly, we explored the difference between 3 types of perceptual misinformation: color, location and replacement. We aimed to find out the possible type of misinformation which would lead to a greater alteration of the individuals' original memory.

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